

System Requirements

Hardware Requirements

Minimum Configuration

- **CPU:** 4 cores (8 cores recommended)
- **RAM:** 16 GB (32 GB recommended)
- **Storage:** 100 GB SSD (500 GB+ for production)
- **Network:** 1 Gbps Ethernet

Production Configuration

- **CPU:** 16+ cores
- **RAM:** 64+ GB
- **Storage:** 1 TB+ NVMe SSD
- **Network:** 10 Gbps Ethernet

Software Requirements

Operating System

- **Linux:** Ubuntu 20.04+, CentOS 8+, or RHEL 8+
- **macOS:** 11.0+ (Big Sur or later) - for development only
- **Windows:** WSL2 with Ubuntu 20.04+ - for development only

Runtime Dependencies

- **Python:** 3.9 or higher
- **Java:** OpenJDK 11 or 17 (for Flink and Kafka)
- **Docker:** 20.10+

- **Docker Compose:** 2.0+

Apache Components

Component	Version	Purpose
Apache Kafka	2.8+	Event streaming platform
Apache Flink	1.14+	Stream processing engine
Apache Zookeeper	3.7+	Coordination service

Database

Component	Version	Purpose
ClickHouse	21.8+	Columnar database for PIT queries

Monitoring Stack

Component	Version	Purpose
Prometheus	2.30+	Metrics collection
Grafana	8.0+	Visualization and dashboards

Python Dependencies

Core packages from `requirements.txt` :

```
apache-flink>=1.14.0
clickhouse-driver>=0.2.3
confluent-kafka>=1.8.0
prometheus-client>=0.12.0
pycryptodome>=3.12.0
polars>=0.14.0
pytest>=7.0.0
```

Network Requirements

Ports

Service	Port	Protocol
Kafka	9092	TCP
Zookeeper	2181	TCP
ClickHouse HTTP	8123	TCP
ClickHouse Native	9000	TCP
Flink JobManager	8081	TCP
Prometheus	9090	TCP
Grafana	3000	TCP

Firewall Rules

Ensure the following ports are open between components: - Kafka brokers ↔ Zookeeper - Flink TaskManagers ↔ JobManager - Application ↔ Kafka brokers - Application ↔ ClickHouse - Prometheus ↔ All services (scraping)

Storage Requirements

Disk Space Allocation

Component	Minimum	Recommended
Kafka data	20 GB	100 GB
Flink checkpoints	10 GB	50 GB
ClickHouse data	50 GB	500 GB
Logs	5 GB	20 GB

Component	Minimum	Recommended
Total	85 GB	670 GB

I/O Performance

- **Kafka:** 500+ MB/s sequential write
- **ClickHouse:** 1000+ MB/s sequential read
- **Flink checkpoints:** 200+ MB/s sequential write

Performance Benchmarks

Throughput Targets

- **Ingestion:** 100,000+ events/second
- **Stream Processing:** 50,000+ events/second
- **PIT Queries:** 1,000+ queries/second

Latency Targets

- **End-to-end Processing:** < 100ms (p95)
- **PIT Query Response:** < 50ms (p95)
- **Checkpoint Duration:** < 30 seconds

Scalability Considerations

Horizontal Scaling

- **Kafka:** Add brokers and increase partitions
- **Flink:** Add TaskManagers
- **ClickHouse:** Use distributed tables with sharding

Vertical Scaling

- Increase Flink parallelism

- Increase ClickHouse memory allocation
- Increase Kafka batch sizes

Cloud Deployment Options

AWS

- **Kafka:** MSK (Managed Streaming for Kafka)
- **Flink:** Kinesis Data Analytics or EMR
- **ClickHouse:** EC2 or ECS with EBS volumes
- **Monitoring:** Managed Prometheus + Grafana

Azure

- **Kafka:** HDInsight or Event Hubs
- **Flink:** HDInsight or Synapse Analytics
- **ClickHouse:** VM Scale Sets
- **Monitoring:** Azure Monitor

GCP

- **Kafka:** Dataproc or Pub/Sub
- **Flink:** Dataflow or Dataproc
- **ClickHouse:** Compute Engine
- **Monitoring:** Cloud Monitoring + Grafana

Container Requirements

Docker Resources

```
# docker-compose.yml resource limits
services:
  kafka:
    deploy:
```

```
resources:
  limits:
    cpus: '2'
    memory: 4G
flink-jobmanager:
  deploy:
    resources:
      limits:
        cpus: '2'
        memory: 4G
clickhouse:
  deploy:
    resources:
      limits:
        cpus: '4'
        memory: 8G
```

Security Requirements

Encryption

- TLS 1.2+ for all network communication
- AES-256 for data at rest
- Secure key management (HSM or KMS)

Authentication

- SASL/SCRAM for Kafka
- ClickHouse user authentication
- Flink dashboard authentication

Authorization

- Role-based access control (RBAC)
- Principle of least privilege
- Audit logging enabled

Compliance Requirements

Data Protection

- GDPR compliance for EU data
- CCPA compliance for California data
- PCI-DSS for payment data (if applicable)

Audit Trail

- All feature accesses logged
- Data lineage tracking
- Immutable audit logs